

LIGHTWEIGHT SAFE HELICOPTER

LSH 00 — PRODUCT FAMILY MASTER SPECIFICATION

Generation 1 Platform Family | Revision 1.0 | Preliminary Controlled Engineering Baseline

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Engineering status statement. This document establishes a common family architecture and planning-level technical baselines. Values marked TARGET are design objectives, not demonstrated performance. Values marked OPEN/TBD require analysis, simulation, test, inspection or certification evidence. This family specification does not establish airworthiness.

1. PURPOSE AND FAMILY DEFINITION

The LSH Product Family is defined as a common safety-oriented rotorcraft platform in which the occupied survival capsule, controlled emergency separation architecture, recovery system, fixed mechanical landing supports, emergency-state management and safety authorization philosophy are shared across multiple aircraft configurations. The family is intended to support controlled evolution from a baseline helicopter into larger, heavier, distributed-rotor and buoyancy-assisted variants.

The family is deliberately organized around a reusable Safety Core rather than around one rotor configuration. Each variant shall preserve the common safety principles unless a formal configuration change establishes an equivalent or superior safety architecture.

2. COMMON DESIGN BASELINE

Parameter	Family baseline
Occupied survival boundary	Dedicated separable survival capsule; final occupied landing boundary after successful separation
Emergency separation	Controlled separation of defined aircraft components from the occupied capsule; automatic and independent manual initiation paths
Separation objective	Nominal controlled transition to capsule-only recovery state; exact timing remains [OPEN] until dynamic analysis establishes a safe envelope
Landing gear	Short, thick, fixed mechanical landing supports/skids; no retractable landing gear in the baseline family
Safety authorization	Independent deterministic safety/interlock layer validates any automatic emergency action
Parachute recovery	Dedicated capsule recovery parachute system; canopy count/type/area allocated by sizing and reliability analysis
Emergency logic	Detect → confirm → assess → authorize → separate → stabilize → deploy → descend → attenuate impact → monitor
Occupant protection	Restraints, energy attenuation, structural survival volume, controlled descent and post-landing protection
Configuration philosophy	Shared interfaces and safety functions; variant-specific sizing and loads are controlled allocations
Evidence status	Conceptual/preliminary; no credit for safety performance until objective evidence is accepted

3. FAMILY VARIANT MASTER MATRIX

ID	Variant	Rotor architecture	Propulsion	Capacity	MTOW target (kg)	Overall envelope target m (L×W×H)	Primary role
LSH-G1	Base 7-seat Safety Helicopter	Single main rotor + tail rotor	2 turboshaft	7	15,000	12.8 × 3.6 × 4.2	Baseline platform
LSH-G2	Large Modular Safety Helicopter	Large single main rotor + tail rotor	2–3 turboshaft	10–12	20,000	15.2 × 4.1 × 4.8	Higher payload / utility

ID	Variant	Rotor architecture	Propulsion	Capacity	MTOW target (kg)	Overall envelope target m (L×W×H)	Primary role
LSH-G3	Heavy Utility / Rescue	Heavy-lift single/tandem study architecture	3–4 propulsion units	14–18	28,000	18.5 × 5.0 × 5.8	Rescue / heavy utility
LSH-MR9	Nine-Rotor Safety Aircraft	Distributed 9-rotor	9 electric/turboshaft-driven units; TBD	8–12	18,000	16.0 × 5.5 × 4.8	Distributed propulsion
LSH-TX	Twin-Rotor Safety Aircraft	Tandem or coaxial study architecture	2 primary rotor systems	10–14	24,000	17.0 × 4.8 × 5.5	Rotor-configuration diversification
LSH-HB	Buoyancy-Assisted Safety Aircraft	Rotorcraft + helium buoyancy envelope	2–4 propulsion units; TBD	8–12	TBD; 18,000 planning study mass	20.0 × 7.0 × 8.0	Exploratory buoyancy branch

Note: envelope and mass values are planning targets selected to establish a coherent family study basis. They are not feasibility evidence. The 15,000 kg G1 planning mass is retained from the existing LSH 00 baseline. ■filecite markers are intentionally not embedded in the PDF; this document is a new controlled family specification.

4.1 LSH-G1 — VARIANT MASTER BASELINE

External geometry: Length 12.8 m; fuselage width 3.6 m; overall height 4.2 m; main rotor diameter target 15.5 m; tail rotor diameter target 3.0 m.

Mass properties: MTOW planning target 15,000 kg; operating empty mass, fuel mass, capsule mass and CG remain [OPEN].

Propulsion: Two turboshaft engines; independent engine monitoring; conventional mechanical/FADEC interfaces to be defined.

Capsule architecture: 7-person separable capsule; 2 crew + 5 occupants; structural survival volume with energy-attenuating floor/seat interface.

Parachute/recovery: Clustered capsule parachute recovery system; preliminary allocation: 3–4 main canopies plus reserve/deployment architecture [OPEN].

Emergency separation sequence: Detect/confirm → inhibit check → arm/authorize → isolate energy/fluid interfaces → release defined outer assemblies → capsule clearance → stabilize → parachute deploy → descent → landing attenuation.

Emergency states: Engine loss; control-system degradation; major fire; severe structural damage; loss of controllability; hazardous external event; combined failures.

Family-common features: fixed short mechanical landing supports; separable occupied capsule; deterministic safety authorization; automatic and independent manual initiation paths; post-action monitoring; rollback/fallback where physically possible; configuration-controlled interfaces.

4.2 LSH-G2 — VARIANT MASTER BASELINE

External geometry: Length 15.2 m; width 4.1 m; height 4.8 m; main rotor diameter target 19.0 m.

Mass properties: MTOW target 20,000 kg; capsule and CG closure [OPEN].

Propulsion: 2–3 turboshaft architecture; final count depends on one-engine-inoperative and mission analysis.

Capsule architecture: 10–12 person enlarged capsule with modular internal layout and reinforced primary load path.

Parachute/recovery: Modular multi-canopy capsule recovery system; preliminary allocation 4–6 main canopies [OPEN].

Emergency separation sequence: Same family sequence; larger separation clearances and higher retained-mass/trajectory loads require dedicated 6-DOF analysis.

Emergency states: Propulsion degradation; fire; structural damage; flight-control degradation; hazardous landing trajectory; multi-system fault.

Family-common features: fixed short mechanical landing supports; separable occupied capsule; deterministic safety authorization; automatic and independent manual initiation paths; post-action monitoring; rollback/fallback where physically possible; configuration-controlled interfaces.

4.3 LSH-G3 — VARIANT MASTER BASELINE

External geometry: Length 18.5 m; width 5.0 m; height 5.8 m; lifting-rotor geometry [OPEN] after heavy-lift analysis.

Mass properties: MTOW target 28,000 kg; mass properties [OPEN].

Propulsion: 3–4 propulsion units as a study range; exact architecture [TBD].

Capsule architecture: 14–18 person heavy-duty capsule; modular passenger/rescue configuration.

Parachute/recovery: Large clustered recovery system; canopy count, reefing and deployment architecture [TBD] by sizing and reliability allocation.

Emergency separation sequence: Staged separation may be required to control asymmetric loads and clearance; exact sequence [OPEN].

Emergency states: Engine/propulsion loss; fire; structural damage; control loss; heavy external-load anomaly; severe weather/terrain emergency.

Family-common features: fixed short mechanical landing supports; separable occupied capsule; deterministic safety authorization; automatic and independent manual initiation paths; post-action monitoring; rollback/fallback where physically possible; configuration-controlled interfaces.

4.4 LSH-MR9 — VARIANT MASTER BASELINE

External geometry: Length 16.0 m; width 5.5 m; height 4.8 m; nine rotor units arranged in a distributed lifting architecture.

Mass properties: MTOW target 18,000 kg; rotor disk area, power and thermal closure [OPEN].

Propulsion: Nine independent propulsion/rotor units; technology may be electric, hybrid or other architecture subject to system study.

Capsule architecture: 8–12 person central separable capsule with symmetric or near-symmetric attachment geometry.

Parachute/recovery: Clustered capsule recovery; 4–6 canopy study range [OPEN], with independent deployment monitoring.

Emergency separation sequence: Emergency authorization → energy isolation → controlled outer-module release → capsule stabilization → recovery deployment.

Emergency states: Single/multiple rotor loss; distributed power fault; battery/engine fire; control network degradation; asymmetric lift; collision/obstacle emergency.

Family-common features: fixed short mechanical landing supports; separable occupied capsule; deterministic safety authorization; automatic and independent manual initiation paths; post-action monitoring; rollback/fallback where physically possible; configuration-controlled interfaces.

4.5 LSH-TX — VARIANT MASTER BASELINE

External geometry: Length 17.0 m; width 4.8 m; height 5.5 m; tandem or coaxial rotor geometry [TBD].

Mass properties: MTOW target 24,000 kg; CG and inertia [OPEN].

Propulsion: Two primary rotor systems; propulsion count [TBD] according to selected architecture.

Capsule architecture: 10–14 person capsule; central load path designed to remain isolated from non-essential aircraft mass after separation.

Parachute/recovery: 4–6 canopy study range [OPEN]; deployment logic shall account for rotor/empennage clearance.

Emergency separation sequence: Family sequence with rotor/drive-system trajectory constraints specific to tandem/coaxial architecture.

Emergency states: Rotor/drive fault; asymmetric thrust; fire; transmission fault; flight-control degradation; structural damage; severe loss of controllability.

Family-common features: fixed short mechanical landing supports; separable occupied capsule; deterministic safety authorization; automatic and independent manual initiation paths; post-action monitoring; rollback/fallback where physically possible; configuration-controlled interfaces.

4.6 LSH-HB — VARIANT MASTER BASELINE

External geometry: Length 20.0 m; width 7.0 m; height 8.0 m planning envelope including buoyancy volume; final envelope [OPEN].

Mass properties: MTOW [TBD]; 18,000 kg retained only as an upper-level study reference, not a feasibility claim.

Propulsion: 2–4 propulsion units [TBD]; buoyancy is supplementary and shall not be credited until buoyancy/drag/stability analysis is complete.

Capsule architecture: 8–12 person separable capsule; protected from buoyancy-envelope hazards and isolated from non-survival systems.

Parachute/recovery: Multi-canopy capsule recovery [OPEN]; interaction with buoyancy envelope and descent stability requires dedicated analysis.

Emergency separation sequence: Isolate buoyancy and propulsion hazards → release non-survival assemblies → establish capsule clearance → stabilize → deploy recovery system.

Emergency states: Propulsion loss; envelope damage; helium loss; fire; control loss; high-wind instability; combined buoyancy/rotor fault.

Family-common features: fixed short mechanical landing supports; separable occupied capsule; deterministic safety authorization; automatic and independent manual initiation paths; post-action monitoring; rollback/fallback where physically possible; configuration-controlled interfaces.

11. COMMON SAFETY ARCHITECTURE

Layer	Function	Failure response
L0	Passive structural/occupant protection	Maintain survival volume; isolate critical loads
L1	Sensing and health monitoring	Validity checks, voting, fault flags, degraded mode
L2	State estimation / emergency classification	Confirm event; avoid single-sensor authorization
L3	Emergency decision logic	Select candidate response; AI/automation does not authorize itself
L4	Deterministic safety kernel	Interlocks, limits, forbidden states, sequencing, authorization
L5	Physical actuation/separation	Fail-safe retention/release architecture; unintended release protection
L6	Capsule stabilization	Attitude/state monitoring after separation
L7	Parachute recovery	Controlled deployment with clearance and deployment-state monitoring
L8	Landing attenuation	Energy absorption and occupant load management
L9	Post-landing survival	Fire, electrical, egress and emergency support provisions

The phrase “separates within seconds” is intentionally not frozen as a universal numerical requirement. Each variant requires a time budget covering detection, confirmation, authorization, actuation, physical clearance, stabilization and recovery deployment. A numerical separation-time requirement shall be established only after dynamic and hazard analyses.

12. COMMON FIXED LANDING SUPPORT ARCHITECTURE

All baseline variants use short, thick, fixed mechanical landing supports/skids instead of retractable landing gear. The design intent is to reduce moving mechanisms, eliminate retract-cycle failure modes, preserve emergency-load paths, and provide a robust ground interface. Exact support dimensions, attachment loads, stroke/energy absorption and ground-clearance requirements remain variant-specific engineering allocations.

13. TECHNICAL PARAMETER SET REQUIRED FOR EACH VARIANT

1. Overall geometry: length, width, height, rotor diameter/spacing, ground clearance, capsule envelope.
2. Mass properties: MTOW, OEW, fuel/energy mass, capsule mass, CG range, inertia tensor.
3. Propulsion: engine/drive type, continuous/emergency power, transient response, redundancy, thermal limits.
4. Rotor/propulsor: disk area, loading, tip-speed limits, control authority, failure cases.
5. Structure: primary load paths, materials/allowables, limit/ultimate loads, fatigue/damage tolerance.
6. Capsule: occupant count, internal volume, mass, CG, attachment points, separation loads, impact attenuation.
7. Separation: initiation paths, interlocks, sequence timing, retained/released masses, trajectories, clearance envelope.
8. Recovery: parachute type, canopy count, area, deployment altitude/airspeed envelope, reefing, redundancy, reliability.
9. Landing: fixed support geometry, energy attenuation, sink-rate design target, ground contact loads.
10. Controls/avionics: sensors, voting, emergency logic, command validation, power, communications, data recording.
11. Safety: FHA, FMEA/FMECA, FTA, common-cause, inadvertent separation, fire, collision and human factors.
12. Environment: temperature, rain, icing, wind, lightning, HIRF/EMI/EMC, vibration and storage.
13. Manufacturing: materials, special processes, assembly, inspection, configuration control and supplier controls.
14. V&V: analysis, simulation, component test, subsystem test, system ground test, flight test and evidence closure.

14. FAMILY INTERFACE CONTROL

The following interfaces shall remain controlled across variants: capsule-to-airframe structural attachments; separation command and inhibit interfaces; electrical/data isolation interfaces; fluid/fuel isolation interfaces; parachute deployment interface; landing-support attachments; safety-kernel command interface; cockpit/manual initiation interface; maintenance and inspection access. Variant-specific interface control documents shall allocate loads, pinouts, command states, physical envelopes and verification methods.

15. VERIFICATION AND MATURITY RULES

No variant shall receive safety credit merely because the common architecture is defined. Required closure includes mass properties, structural loads, separation dynamics, parachute sizing and reliability, landing loads, propulsion feasibility, control authority, failure analysis, unintended-separation probability, environmental qualification and applicable certification means of

compliance. Simulation evidence shall not be represented as flight evidence; conceptual definition shall not be represented as demonstrated performance.

16. FAMILY DEVELOPMENT PRIORITY

Priority 1 — LSH-G1: close the common Safety Core and demonstrate that the architecture can be engineered coherently.

Priority 2 — LSH-G2: establish scalable capsule, separation and fixed-gear interfaces for a larger aircraft.

Priority 3 — LSH-MR9: prove the Safety Core is architecture-independent with distributed propulsion studies.

Priority 4 — LSH-TX: diversify rotor architecture while retaining the same survival boundary.

Priority 5 — LSH-G3: address heavy-mass separation, parachute and landing-energy challenges.

Priority 6 — LSH-HB: retain as exploratory branch until buoyancy, drag, stability and hazard studies justify further development.

17. CONTROLLED STATUS AND IP POSITION

The family architecture, terminology, interface concepts, safety logic, capsule/separation philosophy and variant definitions are engineering development material. They shall not be represented externally as certified, flight-proven or globally novel without the corresponding evidence and legal/prior-art work. Potential IP elements shall be maintained in a controlled register and assessed separately for patent, trade-secret, know-how and contractual protection.

18. BASELINE ACCEPTANCE STATEMENT

This LSH 00 Product Family Master Specification establishes a coherent common baseline for six proposed product variants. It is suitable as a parent document for variant-specific requirements, architecture, analysis, safety, verification, manufacturing and IP documents. It does not establish aircraft feasibility, airworthiness, certification compliance, or demonstrated emergency survival capability.

END OF DOCUMENT — LSH 00-FAM REV 1.0